

YEAR 7 • UNIT 1 • HOMEWORK 4

Roots, and How to Hunt Them

Mathematics 1.6 — Square Roots and Cube Roots • 1.5 — Tests for Divisibility

Name: _____

Class: _____

Date given: _____

Date due: _____

How to do this homework

- Write your answers **on this sheet**, **no calculator**, and everywhere **write the calculation, not just the answer**.
- **Section A** is 1.6 on its own. Start with **A1** — the two lists you write there are the tool the rest of the packet runs on.
- **Section B** puts the two sections **together**: last week's divisibility tests are how you attack a square root that is **not** on your list.

Section A • Mathematics 1.6 — Squares, Cubes, and Back Again

Remember from the lesson

5^2 is said "**five squared**" and means $5 \times 5 = 25$. The small 2 counts **how many fives**, so 5^2 is **not** 5×2 . 5^3 is "**five cubed**": $5 \times 5 \times 5 = 125$.

A **root** runs it **backwards**. Squaring 5 gives 25, so the **square root** of 25 is 5; cubing 5 gives 125, so the **cube root** of 125 is 5. One idea, used twice.

A1 The two lists

(a) Fill in the **twelve square numbers** from your notebook.

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2												

(b) Now the **cube numbers**.

n	1	2	3	4	5	6	10
n^3							

(c) One number is on **both** lists. Which one? _____

A2 Out, and back again

(a) Work out **both** of these, then say which is bigger.

$7^2 = \underline{\hspace{2cm}}$

$7 \times 2 = \underline{\hspace{2cm}}$

Bigger: _____

(b) Now these. Use your lists from A1.

(i) $11^2 = \underline{\hspace{2cm}}$

(iv) $10^3 = \underline{\hspace{2cm}}$

(vii) $\sqrt[3]{27} = \underline{\hspace{2cm}}$

(ii) $12^2 = \underline{\hspace{2cm}}$

(v) $\sqrt{81} = \underline{\hspace{2cm}}$

(viii) $\sqrt[3]{216} = \underline{\hspace{2cm}}$

(iii) $4^3 = \underline{\hspace{2cm}}$

(vi) $\sqrt{144} = \underline{\hspace{2cm}}$

(ix) $\sqrt[3]{64} - \sqrt{100} = \underline{\hspace{2cm}}$

(c) One answer in (b) is a **negative number**. Which one? _____

A3 Say it, then write it**Word help – the English tells you which way to go**

square 9

multiply it by itself: 9^2 . **Out.**

the square of 9

the same thing: $9^2 = 81$.

the square root of 81

the number that was squared: 9. **Back.**

cube 4

$4^3 = 4 \times 4 \times 4$. **Out.**

the cube root of 64

the number that was cubed: 4. **Back.**

a square number

a number that **is** on your first list.

consecutive

one after another, with **nothing missed out**. 6 and 7 are consecutive. So are 36 and 49 – they are consecutive **square** numbers.

Turn each sentence into a calculation, then work it out. **Write the calculation first.**

1. Square 11.

Calculation: _____ Answer: _____

2. Find the square root of 121.

Calculation: _____ Answer: _____

3. Subtract the cube root of 27 from the square of 8.

Calculation: _____ Answer: _____

A4 Trap the root**When the number is not on the list**

45 is not a square number, so $\sqrt{45}$ is not whole. But 45 sits between **36** and **49**, which are on the list – so $\sqrt{45}$ sits between **6** and **7**: two **consecutive** whole numbers.

Fill in the two consecutive whole numbers each root sits between.

(a) _____ $< \sqrt{30} <$ _____

(c) _____ $< \sqrt{110} <$ _____

(b) _____ $< \sqrt{70} <$ _____

(d) _____ $< \sqrt[3]{50} <$ _____

(e) Two **consecutive square numbers** add up to **85**. Which two? _____

Their square roots are _____ and _____

Section B · Maths 1.5 meets 1.6 — Testing for a Root**Why a divisibility test knows something about roots**

A square number is built from a **pair**: $36 = 6 \times 6$. So a factor hiding in one of the pair is hiding in the **other one too** — if **3** divides the number, then $3 \times 3 = 9$ **must divide it as well**. A cube is **three** copies, $8 = 2 \times 2 \times 2$, so a factor found once is really there three times.

Screen 1 3 divides it but 9 does **not** $\Rightarrow$ **not** a square number.

Screen 2 2 divides it but 4 does **not** $\Rightarrow$ **not** a square number.

Screen 3 2 divides it but 8 does **not** $\Rightarrow$ **not** a cube number.

Careful. A screen can only ever say **NO**. Passing it does **not** prove the number is a square.

B1 The screen

Run the tests. Fill in every box, then write **NO** if a screen rules the number out or **maybe** if it survives. If the number is **odd**, put a dash in the 2 and 4 columns — those screens say nothing about an odd number.

Number	Digit sum	by 3?	by 9?	by 2?	by 4?	Square?
132						
441						
162						
1296						
108						

(g) One number in the table survived **every** screen and is **still not** a square number. Which one? _____

In one full sentence, say what that tells you about what these screens can and cannot do.

B2 Now find it — split the number up**Split it into two pieces you already know**

Split the number into $b \times c$, then multiply the root of b by the root of c . **What to split off:** squares 9, 4, 100 (roots 3, 2, 10); cubes 8, 1000 (roots 2, 10).

Worked example. $\sqrt{576}$. The digits add to $5 + 7 + 6 = 18$, so **9 divides it**. $576 \div 9 = 64$. Now $\sqrt{9} = 3$ and $\sqrt{64} = 8$, so $\sqrt{576} = 3 \times 8 = 24$. Check: $24 \times 24 = 576$.

Do the same for each of these. Fill in every column.

Find	Which test works	Split it	The two roots	Answer
$\sqrt{441}$				
$\sqrt{1296}$				
$\sqrt[3]{1728}$				
$\sqrt[3]{8000}$				

B3 The challenge — a number you have already met**5832, from Homework 3**

In Homework 3 you ticked **5832** as dividing by 2, 3, 4, 6, 8 and 9, so it survives **every** screen in B1. It is still **not** a square — but it **is** a cube. Follow the steps.

Step 1. 5832 divides by 8. Do the division. $5832 \div 8 =$ _____

Step 2. Your answer to Step 1 is $9 \times 9 \times 9$, so its cube root is _____

Step 3. $\sqrt[3]{8} =$ _____, so $\sqrt[3]{5832} =$ _____ $\times$ _____ $=$ _____

Step 4. Check it by multiplying your answer by itself twice.

B4 Word problem

The photograph. Mr Bowen lines up **400 students** in a **perfect square**. The photographer climbs onto the roof. How many students are in each row — and which test did you use to split 400 up?